

Luhai Tang

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Education

Worcester Polytechnic Institute (WPI), Worcester, MA

Graduating: May 2027

B.S. Robotics Engineering, Electrical and Computer Engineering Minor

GPA: 3.94/4.00

Work/Experience

Advanced Manufacturing LLC | Manufacturing and Automation Intern - *Windsor, CT*

May 2025 - Jan 2026

- Designed a SOLIDWORKS replica of a THEO MA1-65 laser welder, replicating mass and load distribution within $\pm 5\%$, enabling safe motion testing and eliminating collision risk to \$50k+ equipment.
- Programmed and optimized robotic welding routines in ABB RobotStudio (RAPID, AutoPath), achieving a 90% collision-free path success rate and cutting setup time by 40% compared to manual path teaching.
- Developed custom control logic and programming for CNC, FSAM, and robotic systems, enabling repeatable operation in production.

Advanced Manufacturing LLC | Robotic Welding and Manufacturing Engineer - *Windsor, CT*

May 2026 - Aug 2026

- Programmed and debugged a large-scale \$100k+ KUKA industrial robot for friction stir welding applications.
- Developed 15% more efficient motion and control sequence workflows for complex welding paths in both 2D and 3D space.
- Quality controlled and resolved software motion-control issues through simulation and physical debugging to secure system reliability.
- Robot programming workflows are now implemented at Western New England's Advanced Manufacturing curriculum, supporting hands-on instruction for 50+ students per semester.

WPI Robotics Engineering Department | Robotics Lab Assistant - *Worcester, MA*

Sep 2024 - Present

- Repaired and maintained more than 200 lab robots and other equipment, streamlining laboratory testing and improving reliability.
- Supported 3 robotics lab sections by troubleshooting equipment and ensuring smooth operation of student experiments.
- Developed custom side projects and tools to streamline lab workflows, organization, and equipment management.

Projects

Autonomous Robot Navigation and Mapping System

Mar 2025 - May 2025

- Implemented simultaneous localization and mapping (SLAM) algorithms with LiDAR in ROS, enabling autonomous environment mapping with $> 95\%$ positional accuracy.
- Designed and deployed an A* path planning algorithm to generate collision-free routes, reducing navigation time to target waypoints by $\sim 30\%$ versus baseline methods.
- Developed a pure pursuit controller that improved trajectory accuracy by 20% and enabled smooth, reliable robot motion.

UAV UGV Collaboration - WPI, Draper

Aug 2026 - Present

- Redesigning and building a fleet of UAVs to enable cooperative multi-drone field mapping, improving coverage efficiency over the original single-drone system.
- Real-time map processing on an NVIDIA Jetson for UGV onboard computing, eliminating dependency on ground-station processing.
- Developing UGV navigation logic to autonomously traverse terrain while avoiding pre-designated exclusion regions identified from UAV-generated maps.
- Integrating VSLAM and YOLO models for real-time obstacle detection and localization, improving accuracy with current RTK GPS.

UAV Regulation Interface Platform

Jan 2026 - May 2026

- Engineered a Flask-based regulatory knowledge platform with dynamic routes, modular Jinja templates, and region-specific JSON datasets to organize UAV certification, testing, and compliance information.
- Deployed the production site on Render, supporting 50+ users and enabling future development and full-scale implementation; currently piloted by the Taiwan Testing and Certification Center (ETC) as part of their UAV Compliance workflow integration.

Skills

Programming Languages: Python, C, C++, MATLAB, Java, Linux, Git, RAPID

Robotic Software: ROS, ROS2, ABB Robot Studio, HiperMos Curve

Mechanical and Electrical Design: PCB Design, SOLIDWORKS, KiCAD, LTSpice, MultiSim

Manufacturing: 3D Printing, CNC Programming, FSAM

Languages: English, Chinese Mandarin

Leadership & Activities

Pan Asian Association | VP of External Vendor Collaboration, Secretary, Performance Manager

Jul 2024 - Present

- Organized the Pan Asian Festival/Semi-Formal, achieving a 25% turnout increase with over 2,000 music festival attendees.
- Handled a budget of over \$100k+, coordinating with external performance agencies, and handling \$20k+ artist contracts

Society of Asian Scientists and Engineers | Membership Chair, Events Committee

Sep 2023 - Present

- Developed SASE Roadmaps and organized events, improving member participation at Professional events by 15%
- Awarded 2025 Overall Strongest Chapter, 2026 Most Philanthropic HM at the SASE STEM Connect Conference